

GRADE 9 FULL ACADEMIC SYLLABUS FOCUS

Official Term 1 Scope (Mathematics & Core Sciences)

1. MATHEMATICS (Grade 9 Focus)

Scope: High School Preparatory Mathematics - Term 1 Curriculum

Weighting: 25% of Comprehensive Assessment

- Unit 1.1: Real Numbers, Surds, Scientific Notation and Rational Operations
- Unit 1.2: Linear Equations & Coordinate Geometry (Slope-intercept form, midpoints)
- Unit 1.3: Quadratic Expressions, Polynomial Factorization & Difference of Squares
- Unit 1.4: Right-Angle Trigonometric Ratios (Sine, Cosine, Tangent) & Pythagorean Theorem
- Unit 1.5: Descriptive Statistics, Frequency Distributions & Probability Trees

Key Assessment Focus: Multi-step algebraic manipulation & geometric reasoning

2. PHYSICS (Grade 9 Focus)

Scope: Classical Mechanics & Thermal Physics Foundations

Weighting: 25% of Comprehensive Assessment

- Unit 2.1: Measurements, SI Units, Precision Instruments & Uncertainty Estimation
- Unit 2.2: Kinematics: Displacement, Velocity, Uniform Acceleration & Graphical Analysis
- Unit 2.3: Dynamics & Forces: Newton's Three Laws of Motion, Friction & Terminal Velocity
- Unit 2.4: Work, Energy, Conservation of Mechanical Energy & Power Calculations ($P=W/t$)
- Unit 2.5: Thermal Properties: Kinetic Theory of Matter & Heat Capacity Foundations

Key Assessment Focus: Motion graphs interpretation & vector force diagrams

3. CHEMISTRY (Grade 9 Focus)

Scope: Atomic Structure, Quantitative Stoichiometry & Chemical Transformations

Weighting: 25% of Comprehensive Assessment

- Unit 3.1: Particulate Structure of Matter, Brownian Motion & Phase Changes
- Unit 3.2: Atomic Structure: Protons, Neutrons, Electrons & Periodic Trends
- Unit 3.3: Chemical Bonding: Ionic Transfer, Covalent Sharing & Formula Masses
- Unit 3.4: The Mole Concept: Avogadro Conversions & Gram-Mole Dimensional Analysis
- Unit 3.5: Redox Reactions & Oxidation State Tracking in Elementary Synthesis

Key Assessment Focus: Quantitative mass-to-mole bridge & stoichiometric ratios

4. BIOLOGY (Grade 9 Focus)

Scope: Cellular Biology, Plant Systems & Human Physiology

Weighting: 25% of Comprehensive Assessment

- Unit 4.1: Cell Structure, Organelle Functions, Light Microscopy & Magnification ($I=AM$)
- Unit 4.2: Movement into Cells: Diffusion, Osmosis & Active Membrane Transport
- Unit 4.3: Biological Macromolecules (Carbohydrates, Lipids, Proteins) & Enzyme Kinetics
- Unit 4.4: Plant Physiology: Leaf Structure, Stomata Regulation & Photosynthesis Reactions
- Unit 4.5: Human Organ Systems: Double Circulation, Heart Chambers & Digestive Hydrolysis

Key Assessment Focus: Experimental design, enzyme temperature curves & membrane dynamics

